Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Safe and Efficient Enterprise AI Agentic Workflow Orchestration via Advanced Reinforcement Learning**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (√), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication			
	Medium Category	Small Category	Medium Category	Small Category
	Software	*Software Solutions*		
Concept of Proposed Technology	**Core Concept** This technology develops a production-grade enterprise AI agentic workflow orchestration system that optimizes both safety/compliance and operational efficiency. **Key Functions** 1) Constrained Markov Decision Processes (CMDP) ● Mathematically guarantee compliance (authorization, privacy/PII protection, policy adherence) while learning cost-efficient tool/workflow execution from platform execution traces 2) Enterprise policies ● The system compiles enterprise policies into executable runtime checkers, enabling verifiable constraints rather than best-effort prompts			

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Rationale for Joint R&D Project	**Critical Technical Barriers** Enterprise adoption of agentic AI systems is hindered by two critical barriers: (1) safety/compliance risks that cannot be adequately addressed by prompt engineering alone, and (2) high operational costs from inefficient multi-step workflows. **Why Global Collaboration?** This joint R&D combines academic expertise in safe reinforcement learning algorithms with an industry partner's enterprise orchestration platform (governance controls, execution logs, deployment infrastructure), enabling research to be directly productized and validated on real enterprise workloads.
Key Words	1. Safe Reinforcement Learning 2. Agentic LLM Workflow 3. Enterprise Compliance 4. Constrained MDP 5. AI Safety

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	**R&D Objectives** 1) Constrained Hierarchical Agentic Workflow Policy ● Develop a hierarchical RL policy that learns node/tool/recovery selection for tool-using LLM agents under explicit compliance constraints, treating safety and operational efficiency as co-equal objectives. 2) Policy-to-Checker Compiler ● Create a system that converts natural language enterprise policies into executable runtime checkers, using Information-Theoretic Context Summarization (ITCS) to compress policy documents for low-latency constraint evaluation. 3) Offline Safe RL from Platform Logs ● Design algorithms that use real execution traces (success, failure, and safety events) for safe policy improvement without requiring risky online exploration. 4) Runtime Safety Shield with Auditable Routing ● Implement deterministic safe fallbacks (mask/deny/request approval/human review) with logged evidence and rule attribution for regulatory-grade audit trails. 5) Safe Policy Improvement with Rollout Gates ● Develop conservative update mechanisms and DEV→LIVE release gating to prevent regressions in violation rates or SLA metrics.			
R&D Target Specifications	● Severe Safety Violations: Zero tolerance for PII leaks or unauthorized access ● Total Safety Violation Rate: Minimized violation rate ● Audit Coverage: Full coverage with logged rules & evidence ● Task Performance Retention: High performance integrity relative to baseline ● Token Reduction: Significant reduction in complex workflows ● Latency Reduction: Optimized latency for multi-step tasks			
R&D Plan	**Phase 1: Foundation (Months 1-6)** 1) Define violation events, compliance checkers, and reward/cost signals from platform logs 2) Develop policy-to-checker compiler prototype with ITCS compression 3) Establish evaluation harness with red-team testing (PII injection, jailbreaks, unauthorized access attempts) **Phase 2: Algorithm Development (Months 7-12)** 1) Train baseline policy via imitation learning from successful platform execution traces 2) Implement multi-objective CMDP formulation with dual critics (Reward/Safety Cost/Ops			

	Cost) 3) Develop offline safe RL algorithms with conservative policy updates **Phase 3: Integration & Validation (Months 13-18)** 1) Integrate hierarchical workflow policy with platform's orchestration engine 2) Implement runtime safety shield with auditable routing and logging 3) Conduct offline validation via log replay and simulated adversarial scenarios **Phase 4: Deployment & Optimization (Months 19-24)** 1) DEV environment deployment with automated safety/SLA regression testing 2) Gradual LIVE rollout (A/B testing, partial traffic) with monitoring 3) Iterate on Pareto-efficient trade-offs based on real production feedback 4) Document and package as reusable workflow policy/skill library

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	● Lead the design and development of safe reinforcement learning algorithms and CMDP formulations ● Develop the policy-to-checker compiler and Information-Theoretic Context Summarization (ITCS) framework ● Conduct theoretical analysis and provide mathematical guarantees for safety constraints Design evaluation methodologies and red-team testing protocols ● Publish research findings in top-tier venues ● Provide graduate student researchers and computational resources for algorithm development
Expected R&R of Korean SME Partner	● Provide access to enterprise AI orchestration platform infrastructure ● Supply anonymized execution logs and workflow traces for offline RL training ● Integrate developed algorithms into production orchestration engine ● Conduct deployment testing in DEV and LIVE environments ● Provide governance controls and compliance requirements from real enterprise customers ● Lead commercialization efforts and customer validation ● Handle regulatory compliance for Korean and global markets

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Achievements** 1) Constraint Checker Library + Policy Compiler: Reusable "policy-as-code" layer for enterprise compliance 2) Safe Workflow Policy: Deployable hierarchical orchestrator with provable safety guarantees 3) Safety Scorecard + Governance Dashboard: Standard reporting for risk owners **Commercialization Potential** 1) Direct integration into enterprise AI orchestration platforms with US market validation 2) Pathways to global markets via compliance with US/EU AI regulations (NIST AI RMF, EU AI Act) 3) Addresses key enterprise adoption barrier: quantifiable, auditable AI safety **Industrial/Economic Impact** 1) US Market: Strategic entry point for Korean AI solutions into regulated US industries; Enables "safety-guaranteed agent ops" reducing security/compliance friction 2) Korea Market: Addresses enterprise data governance requirements under evolving AI regulations; mitigates AI talent shortage via reusable, safe workflow automation 3) Global Competitiveness: Establishes Korean partner as a leader in safe Enterprise AI, facilitating exports to North America

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